



Module Title

Cybersecurity and Networking

Internship Assignment Submission

CTF Learning Report

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Role: Cybersecurity Intern

Duration: 3 months

Assignment Submission Date: Thursday, 24th September 2026

Day:15 (Fifteen)

Include your CTF score summary (challenges solved + total points) at the top.

Beginner CTF #1 — Linux & Network Investigation

7 challenges · 150 points · pass at 100 · flags unlock sequentially. Authorized training content only.

← Curriculum

Flags: 7 / 7

Score: 150 / 150

CTF PASSED 🏆

CH1 · Where Am I? (Linux Navigation)

15 pts

Captured +15

You are working inside a student lab:

```
/home/student/jayvik-lab/day15
day15/
|-- notes/
|-- evidence/
|-- reports/
+-- .secret/
```

QUESTION: Which single Linux command displays the directory you are CURRENTLY inside?

Submit the answer as the flag in uppercase: JVL{ANSWER}

CH2 · Hidden in Plain Sight (Hidden Files)

20 pts

Captured +20

A lab folder looks like this:

```
jayvik-ctf/
|-- readme.txt
+-- reports/
+-- report.txt
```

A plain 'ls' shows ONLY readme.txt and reports/. But the evidence for this case is inside a folder whose name starts with a dot.

QUESTION: The clue file says "You found the hidden evidence." — the flag records what this technique finds, in UPPER_SNAKE words: JVL{...}
(Hint from the scenario: hidden ... found.)

WHY THIS ANSWER IS CORRECT

Dot-prefixed entries (.evidence/) are hidden from a normal ls. 'ls -a' reveals them — exactly the 'hidden files found' technique from Days 09-10.

CH3 · Permission Detective

20 pts

Captured +20

A file in the lab shows this permission string in `ls -l` output:

```
-rwxr-x---
```

Convert each group to its numeric value (r=4, w=2, x=1) and submit the three-digit flag: JVL{XYZ}

WHY THIS ANSWER IS CORRECT

rwX=7, r-x=5, ---=0 → 750: owner full access, group read+execute, nobody else.

CH4 · Internet Works, Website Doesn't (DNS)

20 pts

Captured +20

On a Linux host:

```
ping -c 4 8.8.8.8      → works (replies received)
getent hosts example.com → FAILS (name does not resolve)
```

QUESTION: Which ONE network service should be investigated first?

Flag: JVL{<service acronym>}

CH5 · Know Your Services (Ports)

20 pts

Captured +20

Match EVERY port to its common service. All four must be correct before the flag is accepted.

22 → ? 53 → ? 80 → ? 443 → ?

(Names only: SSH / DNS / HTTP / HTTPS – in the right slots.)

WHY THIS ANSWER IS CORRECT

22→SSH · 53→DNS · 80→HTTP · 443→HTTPS — the four defaults every scan report is read with.

CH6 · Read the Scan (Nmap Analyst)

25 pts

Captured +25

STATIC training output – no scanning needed:

```
Nmap scan report for training-host
PORT      STATE    SERVICE
22/tcp    open     ssh
80/tcp    open     http
443/tcp   closed    https
3306/tcp   filtered  mysql
```

Answer ALL four questions, then submit the flag = the two OPEN ports joined by '_':

- 1) Which ports are OPEN? 2) Which services sit on the open ports?
3) Which port is CLOSED? 4) Which port is FILTERED?

WHY THIS ANSWER IS CORRECT

22 and 80 are open (SSH, HTTP); 443 closed (reachable, no listener); 3306 filtered (probes hidden, likely a firewall). The flag pairs the open ports.

CH7 · The Final Case

30 pts

🕒 Captured +30

FICTIONAL training incident — read carefully:

```
Server IP: 192.168.56.20/24    Gateway: 192.168.56.1
Nmap: 22/tcp open ssh · 80/tcp open http · 443/tcp closed https
A file shows: -rw-r-----
```

Answer ALL FOUR parts, then submit the combined flag built from YOUR answers:

JVL{<network>_<svc1>_<svc2>_<perm-numeric>}

Part A network address · B broadcast address · C the TWO exposed services (short names) · D numeric permission.

The flag pattern is: network-octets, port 22, port 80, permission — e.g. JVL{a_b_c_d_e}

WHY THIS ANSWER IS CORRECT

192.168.56.0 (network) · 22 & 80 (open = exposed) · 640 (rw-r-----). Flag = JVL{192_168_56_0_22_80_640} — every part came from a skill taught in Days 06-14.

Finished working through the challenges?

Your Day 15 completion also needs the CTF Learning Report — upload it from the Day 15 lesson page.

Day 15 Lesson

1. Write your CTF LEARNING REPORT

1.1 Answer ALL 8 questions in your own words:

1. Which challenge was easiest, and why?

The easiest challenge was "Where Am I?" because I only needed to use the `pwd` command to find my current working directory. It was a simple Linux command that I had already practiced.

2. Which challenge was hardest, and what made it hard?

The hardest challenge was "Final Investigation" because it required me to use different concepts like IP addresses, ports, network addresses, and permissions.

3. Which Linux commands did you use during the CTF?

`Pwd` - to check current directory

`Ls -a` - to see hidden files

`Ip addr` - to test connectivity

`Nmap` - to check ports and services

4. What did you learn about permissions?

I learned that Linux permissions control what the owner, group, and others can do with a file. For example, `750` gives the owner full permission, the group read and execute permission, and others no permission.

5. What did you learn about DNS?

I learned DNS changes a domain name into an IP address. If the IP works but the website name does not, it can be a DNS problem.

6. What did you learn about ports?

I learned that ports are used by different network services. For example, 22=SSH, 53=DNS, 80=HTTP, and 443=HTTPS.

7. What did you learn from reading Nmap output?

I learned how to identify open, closed, and filtered ports and understand which services are running on an open port.

8. What would you revise before the next CTF?

I would revise Linux commands, permissions, DNS, IP addressing, common ports, and Nmap. I would also practice more CTF questions to improve my problem-solving skills.